INCH-POUND
MIL-V-24336A(SH)
8 July 1988
SUPERSEDING
MIL-V-24336(SHIPS)
20 May 1968
(See 6.7)

MILITARY SPECIFICATION

VALVES, PRESSURE REDUCING, FOR OXYGEN SERVICE

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers self-contained reducing valves for gaseous oxygen service. It does not include regulators used for oxygen storage cylinders and generally associated with welding equipment. These are covered by MIL-R-19180.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

BB-N-411	- Nitrogen, Technical
QQ-N-281	- Nickel-Copper-Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-288	- Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4820
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MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-V-5027 - Valves, Check, Oxygen, High Pressure.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 166 - Standard Specification for Nickel-Chromium-Iron Alloys (UNS N06600 and N06690), Rod, Bar, and Wire. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries of other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Configuration. Emphasis shall be placed on ruggedness, reliability, simplicity, and minimum need for maintenance. Valves shall be constructed to minimize the possibility of initiating ignition in high pressure gaseous oxygen. This shall be accomplished by the following:

(a) Materials, for parts wetted by oxygen, shall have the highest spontaneous ignition temperatures and the lowest impact sensitivities compatible with construction and performance limitations.
(b) Surfaces wetted by oxygen shall not have sharp or thin sectioned protrusions (that is, all parts shall have a high ratio of volume to surface area).
(c) Nonmetallic materials (O-rings, gaskets, and so forth) other than the seating insert shall be well removed from the main flow path.
(d) Valves shall require no lubrication.

3.2.1 Fire containment. In addition, valves shall be constructed to contain an oxygen system fire in the event of ignition anywhere in the system. This shall be accomplished by the following:

(a) The pressure containing envelope shall be made entirely of nickel alloy materials and be sectioned.
(b) Joints for the external seals shall provide an effective barrier to leakage in the event of complete consumption of the non-metallic external sealing elements by providing long, close fitting metal-to-metal leakage or flame paths.

3.3 Materials. Materials shall be as specified in table I and shall be selected for compatibility with high pressure gaseous oxygen and to prevent galling, seizing, or excessive wear between parts. The selection of materials for contacting parts shall be such as to eliminate the need for a lubricant while assuring minimum friction and wear. This list of materials is not intended to be restrictive, providing it is demonstrated that proposed alternate materials are satisfactory. Naval Sea Systems Command (NAVSEA) approval is required for the use of any alternate materials.

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TABLE I. List of materials.

Name of parts	Material
Pressure containing envelope	Monel alloy 400 {QQ-N-281 / QQ-N-288 composition A}
Internal trim (seat, poppet, return spring, and diaphragm (where applicable), wetted by oxygen.	Monel alloy 400 {QQ-N-281 / QQ-N-288 composition A} Inconel alloy 600 (ASTM B 166) Brass
Parts not wetted by oxygen and not comprising part of the pressure containing envelope.	Monel (nickel-copper alloy) Brass Bronze CRES Nickel-aluminum-bronze Copper-nickel alloy
Nonmetallic elements	Materials which are not easily deteriorated or otherwise adversely affected by continued exposure to oxygen and which conform to the requirements of MIL-V-5027.

3.3.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 Construction.

3.4.1 Pressure envelope. The pressure containing envelope shall withstand a proof pressure of 1.5 times the maximum inlet pressure rating specified (see 6.2.1) without external leakage or structural failure and a burst pressure of 3 times the maximum inlet pressure rating. The pressure containing envelope shall be as defined in 3.2.1.

3.4.2 Internal passages. Internal portions of the valve wetted by oxygen shall have smooth surfaces and all edges well rounded. Sharp exterior corners shall be prohibited and interior corners shall incorporate fillets to prevent the retention or entrapment of machining chips and burrs or foreign particles. Pressure lines, including the reduced pressure sensing line, gas dome loading line, and valving element balance line, where applicable, shall be internally ported.

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3.4.3 Internal trim. Internal trim shall be readily replaceable without requiring removal of the valve body from the line. Internal parts in the valve wetted by oxygen shall be massive; that is, shall have a high ratio of volume to surface area. The use of small delicate parts shall be avoided.

3.4.4 Nonmetallic seating insert. When required to meet any tight shutoff requirements specified (see 3.5), a nonmetallic seating insert may be incorporated in either the seat or the disc. The construction and location of the nonmetallic seating insert shall minimize the possibility of ignition under a pressure surge.

3.4.5 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.4.6 Accessibility. The valve shall be accessible for adjustment and repair without removal from the line.

3.4.7 Threads. Threads shall conform to FED-STD-H28. Use of threads wetted by oxygen shall be minimized. Any threads wetted by oxygen shall be of rolled construction or shall be completely chamfered and deburred to prevent the possibility of sharp edges or machining burrs in contact with oxygen. Where necessary, provisions shall be incorporated to prevent the accidental loosening of threaded parts. Pipe threads shall not be used.

3.4.8 Springs. Springs shall be constructed so that they will not be compressed solid during any operation of the valve. Spring ends shall be squared and ground. When removed from the valve and compressed solid, the spring shall not exhibit a permanent set exceeding 0.010 inch per inch of spring length, measured 10 minutes after release of the spring.

3.4.9 End preparation. Unless otherwise specified (see 6.2.1), valves shall be supplied with inline 6-inch nipples welded directly to the valve body or fabricated as an integral part of the valve body. Nipples shall be of the same basic material as the body.

3.5 Performance. Performance requirements (capacity rating, accuracy of regulation, required range of set pressure adjustment, tight shutoff capability, and so forth) shall be as specified in the contract or order (see 6.2.1).

3.5.1 External leakage. There shall be no visible leakage, over a 5-minute period, using either bubble fluid or submerging the valve in water.

3.6 Shock and vibration. Valves shall withstand shock and vibration. When specified (see 6.2.1), shock tests shall be in accordance with grade A, class 1 of MIL-S-901 and vibration tests shall be in accordance with type I of MIL-STD-167-1.

3.7 Marking.

3.7.1 Body markings. Valve bodies shall be permanently marked to show the following information:

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(a) Nominal size.
(b) Pressure rating.
(c) Manufacturer's name or trade-mark.
(d) Flow arrow or "inlet" and "outlet".

3.7.2 Identification plates. An identification plate of corrosion resistant material in accordance with MIL-P-15024 and MIL-P-15024/5 shall be attached to the valve and shall list the following:

(a) Manufacturer's name.
(b) Service (for example: OXYGEN 3000/100 PSIG).
(c) Set pressure limits.
(d) Manufacturer's model or part number and drawing number.
(e) Space for 9-digit CID number.

3.8 Drawings. When specified in the contract or order, drawings shall be prepared (see 6.2.2 and appendix A).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection system. When specified in the contract or order, an inspection system program plan shall be prepared (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. The first oxygen reducing valve of a particular size, configuration and intended application, furnished under a contract or order, shall be subjected to the first article examination and tests specified in table II. When specified in the contract or order, a first article inspection procedure and report shall be prepared (see 6.2.2).

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TABLE II. First article examination and tests.

Examination and tests	Test conditions	Purpose of test	Acceptance criteria
Visual examination	----------	To verify conformance to the requirements of this specification.	Complete conformance.
Proof test	Water, air, nitrogen or oxygen, at 1.5 times the maximum inlet pressure rating, applied to the pressure containing envelope.	To test strength and soundness of pressure containing envelope.	No external leakage, permanent distortion, or structural failure.
Nonmetallic materials compatibility tests	As specified in MIL-V-5027. Test pressure adjusted to be applicable to valve under test.	To verify impact resistance of all nonmetallic parts in oxygen.	No charring or deterioration
Capacity and accuracy of regulation tests	Valve set at max., min., and mid-point of specified set pressure range. Inlet pressure and flow demand varied over specified range. Air or nitrogen may be used in lieu of oxygen for this test.	To verify ability of valve to maintain the downstream pressure within the specified control band over the specified range of operating conditions.	Valve shall meet performance requirements specified (see 6.2.1).
External leakage test	Air, or nitrogen, oxygen at the max. pressure rating applied to inlet, valve set at highest setting.	To check for external leakage.	No visible leakage over a 5-minute period using either bubble fluid or submerging valve in water.
Seat tightness test	As specified (see 6.2.1).	To test for seat tightness.	As specified (see 6.2.1).
Endurance test	100 cycles. Flow varied between max. and lock-up. No lubrication. Dry oxygen. Test to extend over period of at least 1 hour.	To determine endurance of internals, particularly any nonmetallic elements.	Proper functioning and retention of performance capabilities during and after 100 cycles.

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4.4 Quality conformance examination and tests. All oxygen reducing valves shall be subjected to quality conformance examination and tests. These shall consist of the visual examination, proof test, external leakage test and seat tightness test specified in table II. Any discrepancies or failures shall be corrected prior to acceptance.

4.5 Degreasing. Prior to assembly for testing, valves shall be degreased as specified in the contract or order (see 6.2.1).

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 Degreasing. After degreasing and testing as specified in section 4, the valves shall be drained and dried with clean, dry, oil-free air or with nitrogen in accordance with type I, grade A or B of BB-N-411. The valve ends shall be capped and sealed and the valves shall be individually packaged.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be as specified in the contract or order (see 6.2.1).

6. NOTES

6.1 Intended use. The valves are intended for use in gaseous oxygen systems on Naval ships.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When first article is required (see 3.1).
(c) Pressure rating required (see 3.4.1).
(d) End preparation requirements (see 3.4.9).
(e) Maximum capacity required, minimum controllable capacity required, adjustable range of set pressures required, and shutoff capability required (see 3.5 and table II).
(f) Accuracy of regulation required (expressed as a percent of the set pressure). Defined as the maximum band over which the regulated pressure may vary when the valve is set at any pressure within the required range of adjustment and is subjected to any combination of inlet pressure and flow demand, within the limits specified (see 3.5 and table II).
(g) Range of inlet pressures.

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(h) When shock and vibration tests are required (see 3.6).
(i) Any other special performance or construction requirements.
(j) Degreasing requirements (see 4.5).
(k) Level of preservation, packing, and marking required if other than as specified (see 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.8 and appendix A	Drawings, engineering and associated lists	DI-E-7031	----
4.1.2	Inspection system program plan	DI-R-4803	----
4.3	First article inspection procedure	DI-T-4901	----
4.3	First article inspection plan	DI-T-4902	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.2.2 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, Military specifications and standards which have been cleared and listed in DoD 5010.12-L (AMSDL) must be listed on a separate CDRL (DD Form 1423), included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract. Technical content shall include the requirements of appendix B, title "Manual Technical Requirements".

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6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.4.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Subject term (key word) listing.

Gaseous oxygen
Self-contained

6.7 Changes from previous issue. Asterisks are not used in this revision to identity changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4820-N040)

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APPENDIX A

DRAWING TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers technical requirements that shall be included on drawings when required by the contract or order. This appendix is mandatory only when the appropriate data item description is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWINGS

30.1 Drawing content. When required by the contract or order, drawings shall contain the following information:

(a) A sectional assembly of the valve.
(b) Bill of materials listing specification, grade, condition, and any other data required to fully identify the properties of the materials proposed.
(c) End connection detail.
(d) Installation dimensions and clearance dimensions required for disassembly.
(e) Recommended assembly torques, or other equivalent procedures, for all threaded parts.
(f) Reference to previous shock and vibration approval for valve and test report numbers.
(g) Ship identification.
(h) Applicable assembly drawing numbers.
(i) Applicable manual number.
(j) CID(APL) number.
(k) Applicable description.
(l) Valve description.
(m) The set pressure and adjustable range of valve.
(n) Required accuracy of regulation over specified range of operating conditions.
(o) Rated accuracy of regulation over specified range of operating conditions.
(p) Required maximum capacity under specified condition.
(q) Rated maximum capacity under specified conditions.
(r) Fail-open capacity (for purposes of relief valve sizing).

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APPENDIX B

MANUAL TECHNICAL REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers technical requirements that shall be included in technical manuals when required by the contract or order.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications. The following specification forms a part of this appendix to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-15071 - Manuals, Technical: Equipment and Systems, Contents Requirements for.

(Copies of specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

30. MANUALS

30.1 Content requirements. Manuals shall be in accordance with type I of MIL-M-15071. In addition, the following shall be included in the manuals:

(a) The approved engineering drawings for the valve (including certification data sheet). These drawings shall be supplemented by additional illustrations where necessary to adequately illustrate operation and maintenance. These additional illustrations may consist of blowouts, partial or full sections, and so forth, and may eliminate extraneous lines and details to clarify the interaction of parts.

(b) Table listing wrench sizes and assembly torques (or other equivalent procedures) for making up all joints and threaded assemblies.

(c) Instructions to permit overhaul by shipyard or other repair facility. These should include procedures for checking all critical dimensions subject to wear or change and the acceptable dimensional limits, surface finish condition, and so forth. Also, the appropriate procedure (that is, part replacement, correction at repair facility, or repair at manufacturer's facility) which should be followed to correct each case of damage or wear.

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(d) Detailed disassembly and reassembly procedures. In addition to providing procedures for the complete disassembly and reassembly of the equipment, maintenance and troubleshooting sections shall contain, or refer to only the limited disassembly and reassembly required to accomplish each particular operation. This is intended to reduce the possibility of unnecessary disassembly and unnecessary disturbance of adjustments when performing specific or limited maintenance or troubleshooting operations.

(e) Adjustment procedures.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-V-24336A(SH)	VALVES, PRESSURE REDUCING, FOR OXYGEN SERVICE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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